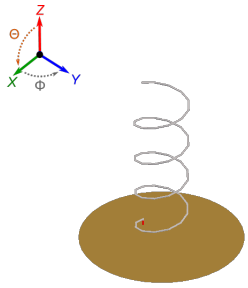
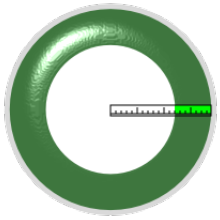
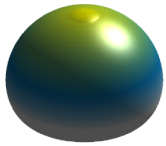
Axial-mode wire helix





Large (> λ/2)



Bidirectional
(unidirectional per polarisation)

Quick Summary

Quantity	Typical	Minimum	Maximum
Polarisation	Circular	-	-
Radiation pattern	End-fire beam	-	-
Gain	14 dBi	9 dBi	19 dBi
Performance bandwidth	1.6:1 (45 %)	-	1.8:1 (55 %)
Complexity	Low	-	-
Impedance	140 Ω	100 Ω	200 Ω
Balun	None required	-	-
Beamwidth	30 °	20 °	56 °

Background

This design is for an axial-mode helical antenna with constant diameter and pitch. It is simple to manufacture, insensitive to dimensional tolerances, and is used for terrestrial and aerospace applications in the HF, VHF, UHF and microwave bands [Kraus and Marhefka].

The antenna radiates a circularly polarized end-fire beam in the direction of the helix axis and the input impedance is nominally resistive over the operating band.

Physical Description

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The helical conductor can be self-supporting if sufficiently rigid wire or tubing is used. Otherwise radial insulating struts mounted on an axial rod can provide support or a Styrofoam (or any low-loss, low relative permittivity material) former can be used to seat the conductor and provide mechanical stability [Kraus and Marhefka, Section 8.3].

If preferred the round conductor can be replaced by a strip of width w=2d, with d the wire diameter [Elliott, pp. 324-326].

Feed Method

The bottom end of the helical conductor is soldered to the centre pin of a coaxial connector mounted under the ground plane.

The feed pin is located on the periphery of the cylinder. The swept frequency input reflection coefficient of the peripherally fed helical antenna is superior to an axially fed helix if a well designed impedance taper is introduced between the connector pin and the start of the uniform helix [Kraus and Marhefka, Section 8.3], [Baker, 1980], [Baker, 1985].

The diameter of the ground plane is typically about 0.75 λ to reduce back-lobe radiation and enhance forward gain. The circular ground plane can be cupped, or a square plane or grid reflector with aperture dimensions of about 0.1 λ by 0.1 λ [Kraus and Marhefka, Section 8.3].

Operation Mechanism

The uniform axial mode helical antenna has between 3 to 5 turns per wavelength and circumference approximately one wavelength at the center frequency. The corresponding pitch angle is between 11 and 18 degrees.

A right/left handed helix transmits and receives right/left hand circularly polarized radiation as defined by the IEEE [Kraus and Marhefka, p. 46 and Section 8.11].

The N-turn axial-mode helix supports a travelling wave current distribution with exponential amplitude decay which accounts for radiation loss. It can be modeled as an end-fire array of N identical single turn travelling wave loop antennas, each of circumference one wavelength, spaced by the pitch distance and circularly polarized [Kraus and Marhefka, Sections 8.1 and 8.8], [Elliott, Sections 2.7 and 2.8].

Performance

Impedance Characteristics

The input impedance exhibits oscillatory behavior at the high-frequency end of the band as the input resistance decreases from 200 to 100 Ω.

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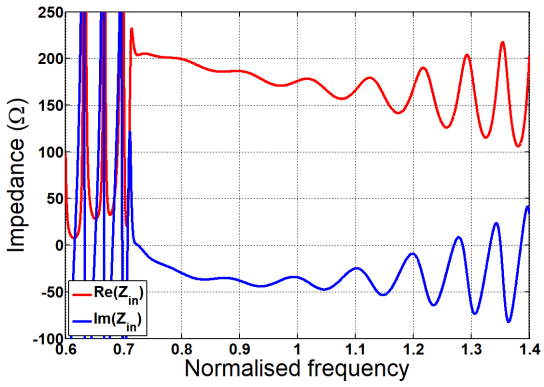
The capacitive reactance can be reduced by decreasing the length of the coaxial feed-pin, and the match can be further improved by an impedance taper as mentioned above.



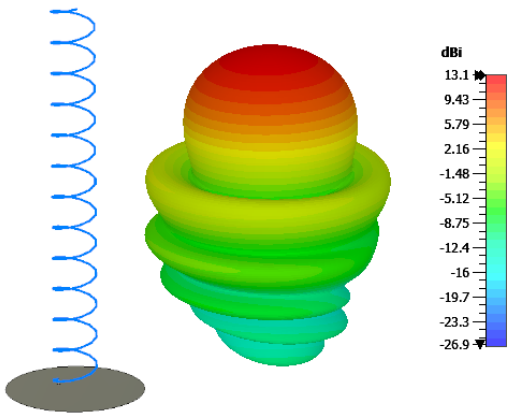
Typical input impedance

Radiation Characteristics

The plots shown below are for a 12.4 turn helix. The gain lies between 10 and 15 dB for 55% bandwidth.

The pattern is stable, with well defined end-fire lobe low side and back lobes over the operating band and slight main-beam pattern asymmetry about the axis.

If required the axial ratio can be reduced by terminating the helix with a planar Archimedes spiral [Kraus and Marhefka, Section 8.23], [Baker, 1985].



Typical polarisation specific gain radiation pattern at the centre frequency

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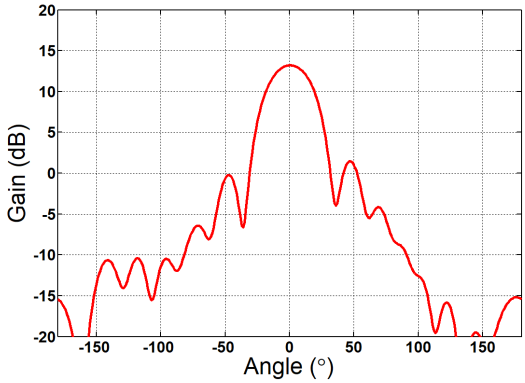
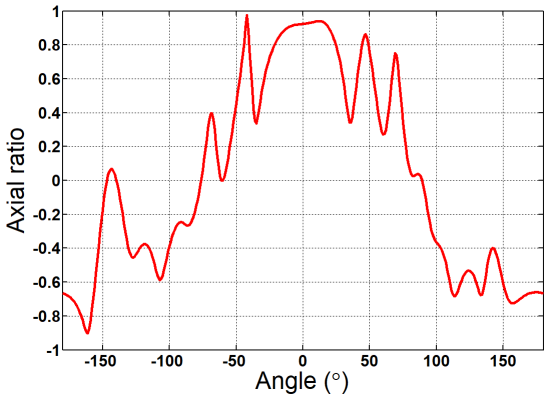
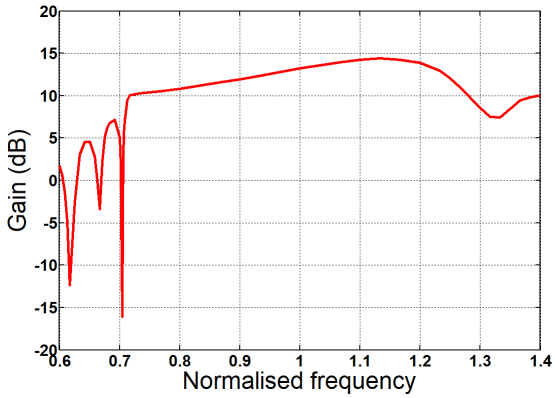


Typical circularly polarised radiation pattern



Typical axial ratio at the centre frequency



Typical boresight gain versus frequency

References

J.D. Kraus and R.J. Marhefka, Antennas for all applications, McGraw-Hill, 2002, Sections 8.1-8.4, 8.8, 8.10-8.14, 8.20. (Kraus invented the axial-mode helical antenna in 1946. They give a comprehensive and practical review of the extensive work done by Kraus and his collaborators since then.)

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H.E. King, J.L. Wong and E.H. Newman, Helical antennas. In Antenna engineering handbook, 4th Ed, J. Volakis (Ed), McGraw-Hill, 2007, Sections 12.1-12.3. (This is concise summary of the important work by King and Wong on uniform and non-uniform axial-mode helical antennas.)

H. E. King and J. L. Wong, “Characteristics of 1 to 8 wavelength uniform helical antennas,” IEEE Transactions on Antennas and Propagation, vol. AP-28, no.2, March 1980, pp. 291-296.

J. L. Wong and H. E. King, “Broadband quasi-taper helical antennas,” IEEE Transactions on Antennas and Propagation, vol. AP-27, no.1, January 1979, pp. 72-78.

R.S. Elliott, Antenna theory and design, Revised Ed., IEEE Press, 2003.

D.E. Baker, “Design of a broadband impedance matching section for peripherally fed helical antennas,” Antenna Applications Symposium, Allerton, University of Illinois, September 1980. (Shows downward shift of cutoff-frequency for helix supported by PVC tube instead of a Styrofoam cylinder.)

D.E. Baker, “Measured performance of a broadband matching section for peripherally fed helical antennas,” Transactions of the South African Institute of Electrical Engineers, vol. 76, no. 2, June 1985, pp. 56-61. (Shows Smith chart impedance plots; effect of number of turns on S11 magnitude; and effect of planar Archimedes spiral end-termination on axial ratio.)

Model Information (FEKO)

Model 1

Thin-wire helix model on a finite ground plane.

The model uses a finite ground plane and a thin-wire approximation of the helix. All parts are PEC.

Model Information (CST Studio Suite)

Model 1

Thin wire model on a finite ground plane with a coaxial feed.

This model uses a thin-wire approximation of the helix and a finite ground plane. It is fed using a coaxial feed line and is to be solved using the I-solver.

Variation 1

A variation of the original export model with an added cylindrical dielectric cover.

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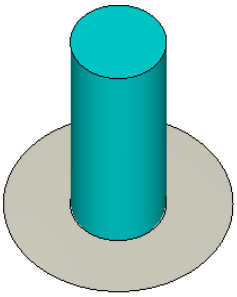
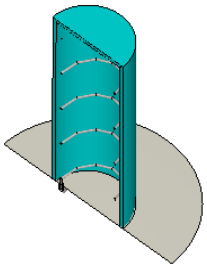
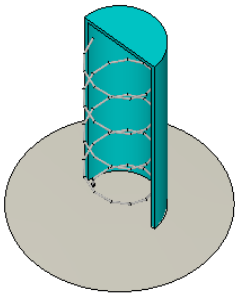
The variation includes a cylindrical dielectric cover as part of the export model. A number of special cylinder parameters have been added to the model, derived from parameters of the original export model. The preferred solver is the I-Solver.







3D View of the variation (full view, cross-sectional view and split view)

Model Validation

The models have been checked against measured radiation patterns [King, Wong and Newman], [King and Wong].

The input impedance simulated using the various codes have been compared.

Each export model has been validated to give the expected results for several parameter variations in the design space.

Magus Analysis

The internal performance estimation is expected to be similar to a full 3D-EM analysis. Expect:

- Small frequency offsets (-3% to +3%)
- Possibly inaccurate reflection coefficients below -15 dB
- Radiation patterns outside of the main beam may be slightly inaccurate

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Design Guidelines

The Antenna Magus design is limited by the maximum allowable size of the models.

- To increase the gain, increase the number of turns

- A ground plane with cylindrical or conical cup can be used to improve the front-to-back ratio.

- A shim with low pitch angle at the bottom of the helix, or else use an impedance taper can be used to improve the impedance match [Kraus and Marhefka, Sections 8.3 and 8.23], [Baker, 1980], Baker, 1985].

- Further improvements are obtained by terminating the helix with a flat Archimedes spiral [Kraus and Marhefka, Sections 8.3 and 8.23], [King, Wong and Newman], [King and Wong], [Baker, 1980], Baker, 1985].

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